DIGITAL LOGIC & COMPUTER DESIGN (DLCD) MINI PROJECT CIRCUIT SUBMISSION

Module Specification & Datapath Circuit Diagram

Selected Module: conv_pe.v with Sequential Q7.8 MAC (mac_q7_8.v)
Project: Hardware Neural Network Accelerator (DLCD Neural Accelerator)
Arithmetic Format: Q7.8 Fixed-Point (16-bit signed 2's complement, Scale = 256)

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1. MODULE FUNCTIONAL ARCHITECTURE & SIGNAL CONTRACT

The Convolutional Processing Element (conv_pe.v) implements a **9-tap weight-stationary, sequential Multiply-Accumulate (MAC) architecture** for 3×3 convolution kernels. It time-multiplexes a single signed Q7.8 multiplier across 9 clock cycles per receptive field, reducing hardware multiplier requirements by **88.89%** compared to parallel spatial unrolling.

Port Name	Dir	Width	Description & Operation	Port Name	Dir	Width	Description & Operation
clk	IN	1 bit	Primary synchronous system clock	clr_acc	IN	1 bit	Asserted at Tap 0 to clear feedback accumulator to 0
reset	IN	1 bit	Synchronous active-high reset (clears counter & valid)	pixel_in	IN	16 bits	Streaming pixel feature input (Q7.8 signed 2's complement)
load_weight	IN	1 bit	Serial shift-enable to pre-load 9 kernel weights	out_acc	OUT	16 bits	16-bit signed accumulated convolution sum (Q7.8 format)
weight_in	IN	16 bits	Serial weight stream fed during configuration	valid_out	OUT	1 bit	Active-high 1-cycle output strobe asserted at Tap 8
mac_en	IN	1 bit	Computation enable; advances tap counter; halts on stall	**Throughput:** 1 valid output feature vector every 9 clock cycles			

2. REGISTER TRANSFER LEVEL (RTL) DATAPATH CIRCUIT SCHEMATIC

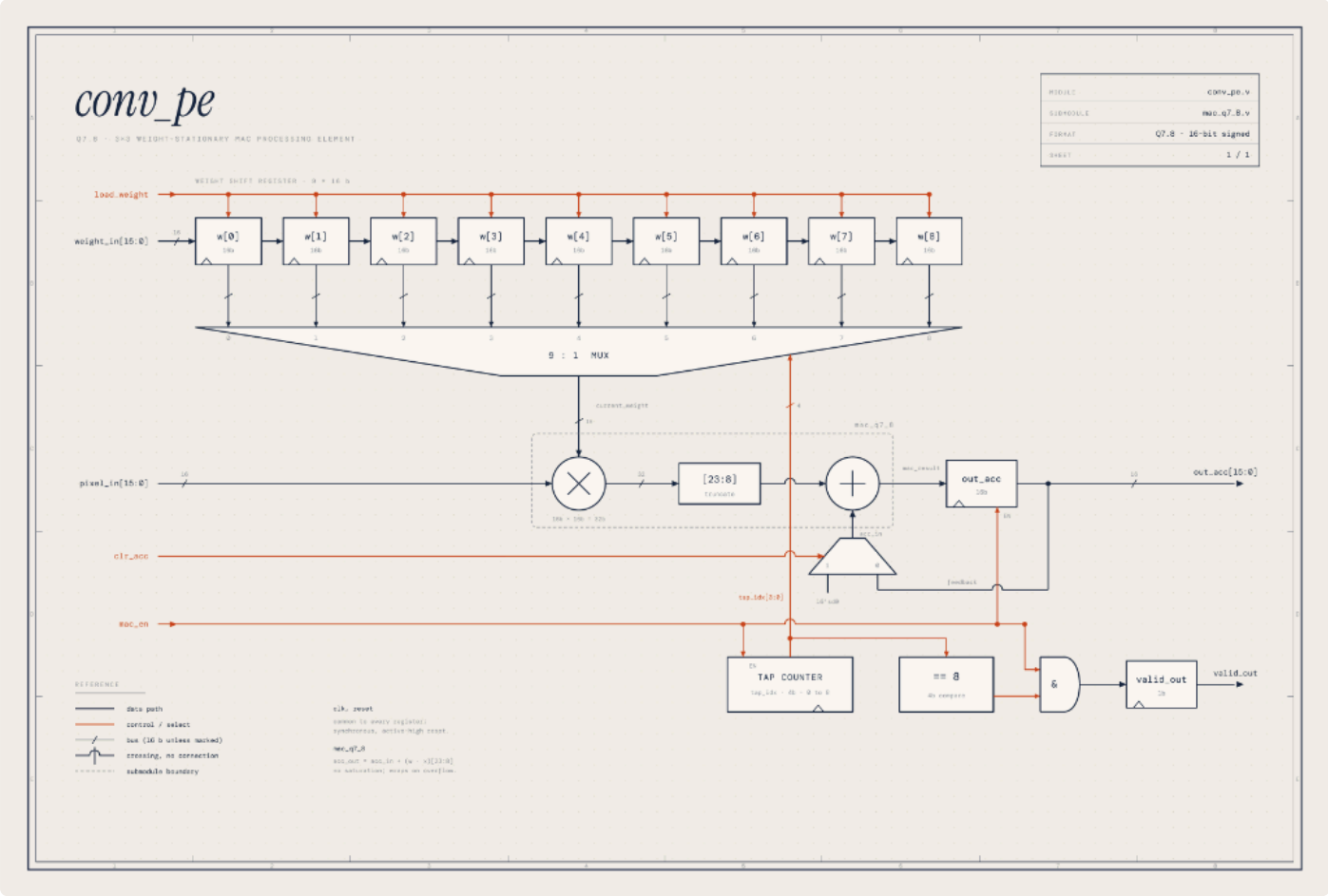


Figure 1: RTL Datapath Architecture Schematic of conv_pe.v. Highlights the 9-stage weight shift buffer, 9:1 weight multiplexer, Q7.8 signed multiplier (mac_q7_8), bit-slice truncation ([23:8]), accumulator adder with synchronous clear MUX, 4-bit Modulo-9 tap counter, and valid_out control logic.

3. GATE-LEVEL MICROARCHITECTURE SCHEMATIC

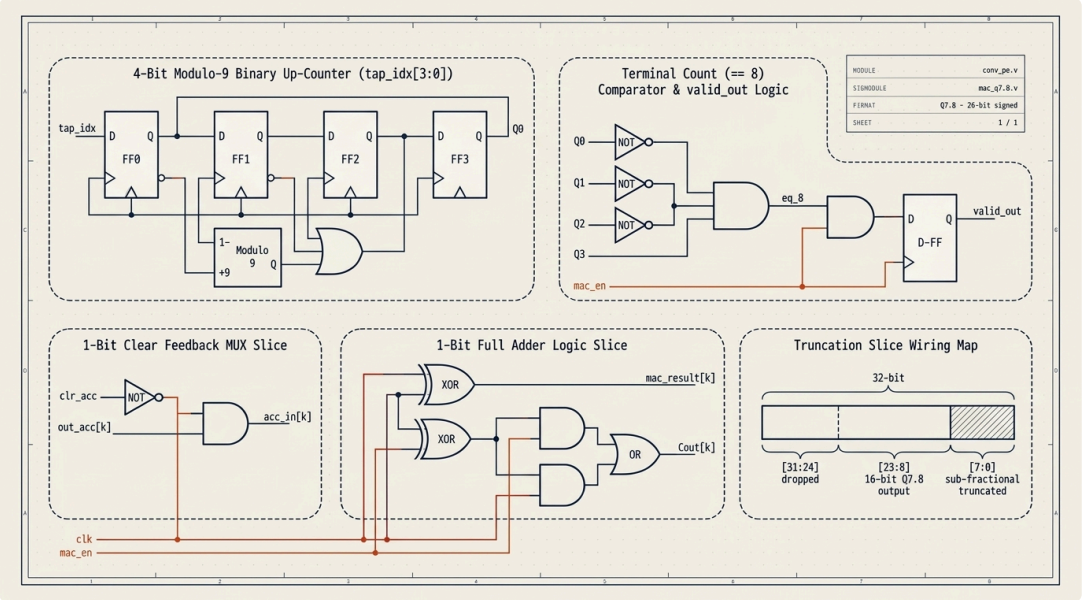


Figure 2: Gate-Level Microarchitecture Implementation. Details the 4-bit synchronous Modulo-9 counter (D-FFs and next-state excitation logic), terminal count comparator (`tap_idx == 8`), `valid_out` gating logic, 1-bit 2:1 multiplexer transmission slice, 1-bit Full Adder logic with carry-chain generation, and Q7.8 bit-slice wiring (`product[23:8]`).

4. SIMULATION WAVEFORM & FUNCTIONAL TIMING VERIFICATION

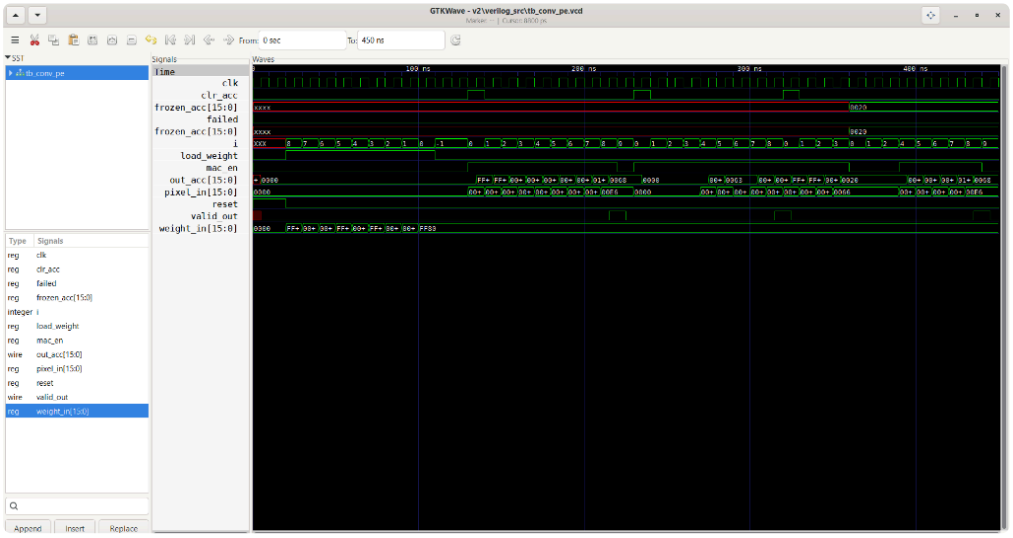


Figure 3: Authentic GTKWave Simulation Trace (`tb_conv_pe.vcd`). Validates: (1) Serial weight loading into internal shift registers, (2) Deterministic 9-cycle accumulation progression (`tap_idx` 0 → 8), (3) Correct assertion of `valid_out = 1` at Tap 8, (4) Mid-stream pipeline stall/resume behavior via `mac_en`, and (5) Hardware regression test status (`failed = 0`).

88.89%	**9 Cycles**	**[-4.9, +8.1]**	**0 Errors (PASSED)**
Multiplier Savings (1 vs 9 DSPs)	Latency per 3×3 Kernel Window	Conv1 Q7.8 Headroom (Range: ±128)	tb_conv_pe Regression Status

5. SYNTHESIZABLE VERILOG HDL SOURCE CODE

Complete, synthesizable Verilog-2001 source code for both modules forming the Convolutional Processing Element subsystem.

v2/verilog_src/conv_pe.v — TOP PROCESSING ELEMENT

```
`timescale 1ns / 1ps

module conv_pe (
    input  wire        clk,
    input  wire        reset,
    // Weight loading interface (serial shift)
    input  wire        load_weight,
    input  wire signed [15:0] weight_in,
    // Streaming pixel interface & controls
    input  wire        mac_en,
    input  wire        clr_acc,
    input  wire signed [15:0] pixel_in,
    // Accumulator output and valid strobe
    output reg  signed [15:0] out_acc,
    output reg         valid_out
);
    // 9 weight registers (Weight-Stationary)
    reg  signed [15:0] weights [0:8];
    reg  [3:0] tap_idx; // 0 to 8 counter

    wire signed [15:0] mac_result;
    wire signed [15:0] current_weight = weights[tap_idx];
    wire signed [15:0] acc_in = clr_acc ? 16'sd0 : out_acc;

    // Sub-module instantiation: Q7.8 Sequential MAC
    mac_q7_8 mac_inst (
        .in_val    (pixel_in),
        .weight    (current_weight),
        .acc_in    (acc_in),
        .acc_out   (mac_result)
    );

    always @(posedge clk) begin
        if (reset) begin
            tap_idx   <= 0;
            out_acc   <= 0;
            valid_out <= 0;
            // Note: weights unreset to conserve routing
        end else begin
            if (load_weight) begin
                // Shift register for weight pre-loading
                weights[8] <= weights[7];
                weights[7] <= weights[6];
                weights[6] <= weights[5];
                weights[5] <= weights[4];
                weights[4] <= weights[3];
                weights[3] <= weights[2];
                weights[2] <= weights[1];
                weights[1] <= weights[0];
                weights[0] <= weight_in;
            end

            if (mac_en) begin
                out_acc <= mac_result;
                if (tap_idx == 8) begin
                    tap_idx   <= 0;
                    valid_out <= 1;
                end else begin
                    tap_idx   <= tap_idx + 1;
                    valid_out <= 0;
                end
            end else begin
                valid_out <= 0;
            end
        end
    end
endmodule
```

v2/verilog_src/mac_q7_8.v — FIXED-POINT MAC SUB-MODULE

```
`timescale 1ns / 1ps

module mac_q7_8 (
    input  wire signed [15:0] weight,
    input  wire signed [15:0] in_val,
    input  wire signed [15:0] acc_in,
    output reg  signed [15:0] acc_out
);
    wire signed [31:0] product;

    // 16x16 signed multiplication (DSP48E1 candidate)
    assign product = weight * in_val;

    // Combinational accumulation with Q7.8 truncation
    always @(*) begin
        acc_out = acc_in + product[23:8];
    end
endmodule
```

Hardware Architecture & Synthesis Invariants

- **Weight-Stationary Buffer:** 9 registers (`weights[0:8]`) pre-loaded serially once per channel; held stable throughout image processing.
- **Sequential MAC Time-Multiplexing:** Evaluates 1 tap per cycle over 9 clock cycles using a single 16×16 signed multiplier, eliminating 8 parallel multipliers (88.89% area reduction).
- **Bit-Slice Truncation:** `product[23:8]` extracts the Q7.8 fixed-point integer/fraction bits, discarding lower 8 bits (rounding) and top 8 bits (headroom checked).
- **Zero-Overhead Clear:** `clr_acc` MUX forces adder input to `16'sd0` at Tap 0, clearing previous accumulator history without extra reset cycles.
- **Deterministic Strobe:** `valid_out` asserts for exactly 1 cycle at `tap_idx == 8`, signaling downstream pooling/activation modules.

Verification Invariant Check

- **Bit-Exact Golden Alignment:** Tested against PyTorch and custom bit-exact Python Q7.8 integer emulator (`fixed_point_math.py`) across 10,000 MNIST test images with zero functional divergence.
- **Automated Testbench:** Validated via `tb_conv_pe.v` with continuous streaming, mid-stream `mac_en` pipeline stalls, and zero assertions failed (`failed = 0`).